URCHIN – Underwater Robotic Construction Modules Using High-Tolerance Interlocking Nodes

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Regular Paper

Introduction

Underwater construction is hazardous and resource-intensive [1], [2]. Robotic automation enables safer, scalable assembly, but subsea manipulation suffers from high spatial uncertainty. We propose **URCHIN**: a modular voxel system using specialized geometry and passive magnetic alignment to overcome marine manipulation errors. Key contributions:

- **Self-Aligning Interface:** Magnetostatic model predicts inter-voxel pull forces for passive snapping without high-precision sensing.
- **Manipulator-Agnostic:** Grasp-optimized geometry fits standard commercial parallel-jaw grippers.
- **Localization-Ready:** Integrated slots for fiducial tracking markers.

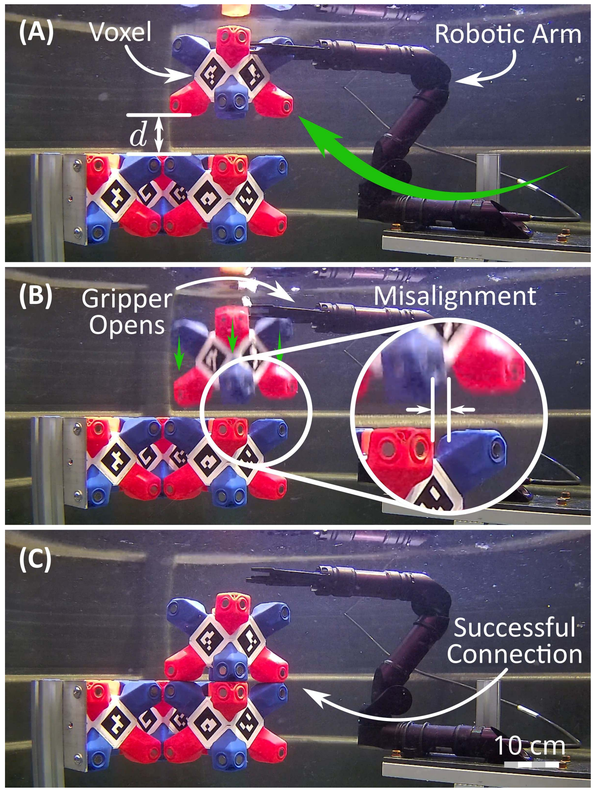


Figure 1: (A) A REACH Robotics Alpha Arm moves an URCHIN voxel into position. (B) The released voxel drops into initial misalignment. (C) Passive magnetic attraction aligns and snaps the voxel into place.

Architecture

URCHIN is designed around the following core design principals:

- **Parallel-Jaw Compatibility:** Flat-parallel faces fit standard commercial grippers.
- **Passive Magnetic Alignment:** N52 neodymium magnets draw modules together across positional offsets.
- **Shear Resistance:** Interlocking male-female node geometry transfers off-axis shear loads directly through structural features.
- **Buoyancy Mitigation:** Internal gyroid infill vents allow complete water flooding during submersion.
- **Structural Rigidity:** Internal aluminum hexagonal braces lock shell halves against bending and torsion.

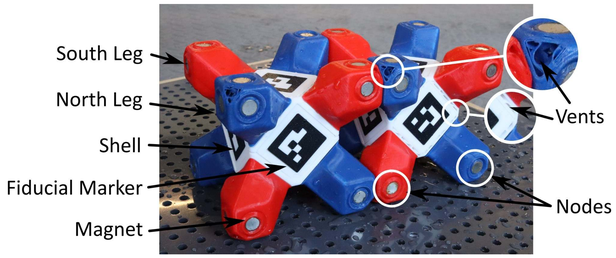


Figure 2: URCHIN voxel component overview and defining features.

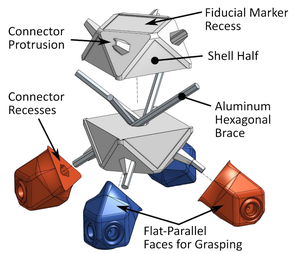


Figure 3: Exploded view showing internal structure and leg geometry.

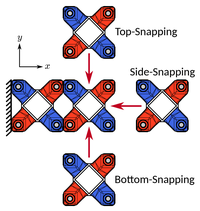


Figure 4: Evaluated multi-directional snapping vectors.

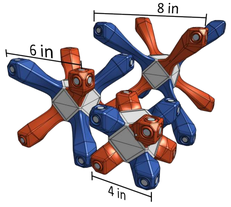


Figure 5: Scalable module geometry (4 in, 6 in, and 8 in prototypes).

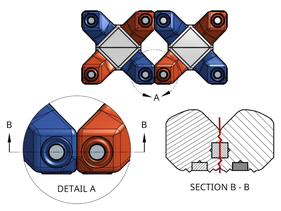


Figure 6: Section view showing interlocking anti-shear node geometry.

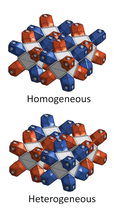


Figure 7: Magnetic leg configurations.

Modeling

VOXEL-TO-VOXEL INTERACTION MODEL

A reciprocal cubic model captures near-field ($\frac{1}{d^2}$) and far-field ($\frac{1}{d^4}$) inter-voxel pull force $F_{\text{est}(d)}$ as a function of separation distance d:

$$F_{\text{est}}(d) = \frac{aK}{bd^3 + cd^2 + ed + f} \tag{1}$$

where $K = \pi B^2 \frac{R^4}{\mu}$. Empirical parameters (a, b, c, e, f) are solved via nonlinear optimization.

ANALYTICAL BENDING MODEL

Max allowable external point load F_{applied} at the tip of an N-voxel cantilevered chain before joint detachment:

$$F_{\text{applied}} \leq \frac{\frac{l}{L}F_{\text{ref}} - \frac{1}{2}N^2W}{N - \frac{1}{2}} \tag{2}$$

where F_{ref} is contact pull force, l is moment arm, L is voxel length, and W is submerged weight.

Experiments & Results

SNAPPING DISTANCES

Evaluated snapping limits using a 4-DOF REACH Alpha Arm. Demonstrated reliable passive alignment underwater up to:

- **Top-Down Offset:** 4.44 cm (1.75 in)
- **Side-In Offset:** 1.90 cm (0.75 in)
- **Bottom-Up Offset:** 1.51 cm (0.59 in)

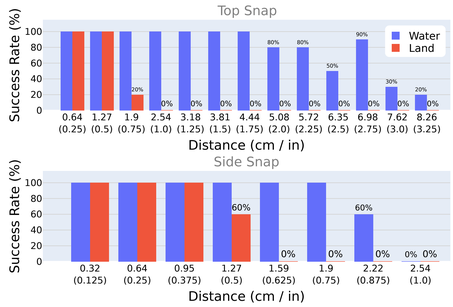


Figure 8: Snap success rates for top-down and side-in configurations underwater vs. land (10 trials per offset).

PULL FORCE VERIFICATION

Used a tensile testing system to determine:

- **Mean Contact Separation Force:** 64.8 ± 5.95 N.
- **Model Accuracy:** Reciprocal cubic model achieved an **RMSE of 0.118 N**—a **20x error reduction** over standard Zurek models [3].

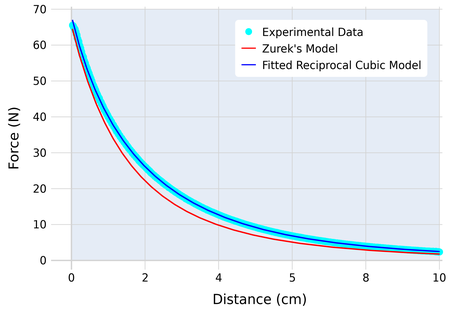


Figure 9: Measured magnet pull force between voxel faces vs. Zurek's model [3] and fitted reciprocal cubic model (Equation 1).

BENDING FAILURE

Tested submerged cantilevered chains ($N = 1$ to 4) using an external pulley rig. Experimental detachment loads matched predictions by Equation 2 within **14.90% average error** across all chain lengths.

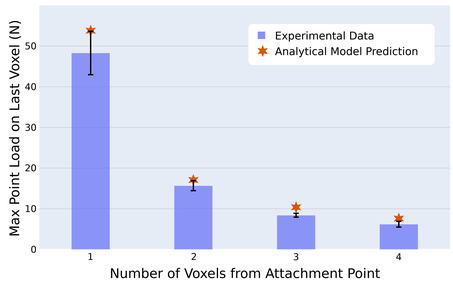


Figure 10: Maximum point load vs. distance from fixed end for cantilevered voxel chains.

AUTONOMOUS 1D/2D/3D ASSEMBLY

Executed unassisted placement in 1D, 2D corner, and 3D vertex configurations under initial misalignments. Successfully assembled complex structures (arches, overhangs, branching chains).

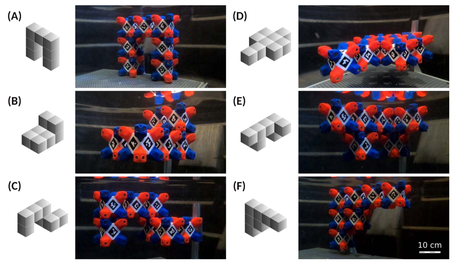


Figure 11: Examples of structures autonomously assembled from URCHIN voxels.

Conclusion

- Passive magnetic alignment successfully mitigates spatial uncertainty in underwater construction.
- **Future Work:** Integration of passive mechanical interlocks for added shear resistance and testing in dynamic marine environments.

Acknowledgements

This work was supported in part by ONR awards N00014-22-1-2114 & N00014-23-1-2171, and NSF awards 2349747 & 2437138. We thank Christopher Holm and Brian Van Stratum for design feedback, and the Robotic Decision Making Laboratory (RDML) for support.

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